

Design of a Medical Expert System to Consult Selecting an Examination Department and to Provide the Path Signal for a Mobile Robot to Lead Patients to the Consulted Department

Ngoc Anh Mai

Le Quy Don Technical University
Hanoi, Vietnam
maingocanh@lqdtu.edu.vn

Hoang Thi Kieu Ly

Banking Academy of Vietnam
Hanoi, Vietnam
lyhtk@hvn.edu.vn

Manh Nguyen Van

Le Quy Don Technical University
Hanoi, Vietnam
vanmanh.isi@lqdtu.edu.vn

Thi Ngoc Mai Nguyen*

Le Quy Don Technical University
Hanoi, Vietnam
ngocmai@lqdtu.edu.vn

Abstract—The article introduces the design of a medical expert system consulting a medical examination department and a solution of applying the consultation results on a mobile robot to lead patients to the consulting department. This research will be experimented on a mobile robot placed in a simulated examination department environment with 5 typical examination departments. To consult the department according to symptoms, the Medical Expert System will use 15 pathological signs and 2 test results to make diagnostic results for 4 common diseases. Experimented results showed that Medical Expert System can consult selecting the examination department based on the disease symptoms that the patient provided and the result of examination department selection is encoded into the path signal for the mobile robot.

Keywords—Medical expert system, department consultation, mobile robot

I. INTRODUCTION

According to [1], Medical Expert System (MES) is one kind of expert systems (a branch of Artificial Intelligence) imitating knowledge of physician experts to deal with pathological problems. For this reason, MES has a high potential to be widely applied in many aspects of health care. Compared to other traditional medical programs, MES has to transform complex knowledge bases of experts into simpler knowledge bases for programming to help a computer think like a doctor. Numerous MESs have assisted physicians in advising and making decisions about various pathologies such as [2-5].

Mobile robot has the movement ability to the assigned position based on signals from sensors [6]. Mobile robots can flexibly travel in unknown environment without the information of structure and/or objects. Alternatively, mobile robots can follow a pre-defined navigation route such as a line-based map to approach a target. Humanoid mobile robot is one kind of mobile robots shaped like human with basic components such as head, neck, torso and arms [7]. Normally, humanoid robots are equipped with the ability to observe the surrounding environment using cameras as well as avoiding obstacles using sensors acting as human senses such as hearing

seeing and touching. This is the reason why a humanoid mobile robot shaped like a nurse is able to friendly work in a hospital area. Nurse mobiles has been widely applied in hospitals to accomplish different tasks related to support doctors, hospital staffs and patients.

Robots in the medical field are being researched with the support of modern technology that allows minimized human-to-human contact and ensures cleaning and disinfection in hospitals and medical facilities. Medical robots also play an active role in managing infections in hospitals, especially during the COVID-19 pandemic. Currently, there are many different types of robots that have been developed and applied in the field of healthcare, such as Nursing robots [8], Reception Robots [9], Remote online health consulting robots [10], Surgical robots [11],...

In this study, a humanoid robot is designed with two arms for holding a computer containing MES that allows patients to interact to the nurse robot to provide their disease symptoms. Based on the provided symptoms, the nurse robot will infer to the suitable examination department(s) and then guide the patient to the department along road markings. The suitable examination department play the role of a target for robot to solve the global path planning. During moving from the standing position to the target, the nurse robot has to travel point-to-point along a line-marked map on the road. The basic contents in this paper is organized as follows: i) Design a MES capable of advising on the examination department selection based on the symptoms provided by the patient; ii) Use the consulting results as a signal to choose the route based on the road markings for the mobile robot to guide patients to the suitable examination department.

This article includes: Part 1 introduces the structure and program design of MES; Part 2 describes the application of consulting results to guide a mobile robot; Part 3 presents tests in a simulated and analytical clinic workspace to evaluate the possibility of combining MES and robotic mobile control systems. The conclusion recaps the results of this study and the next direction for further development.

II. MEDICAL EXPERT SYSTEM

A. Structure and operation process of a medical expert system

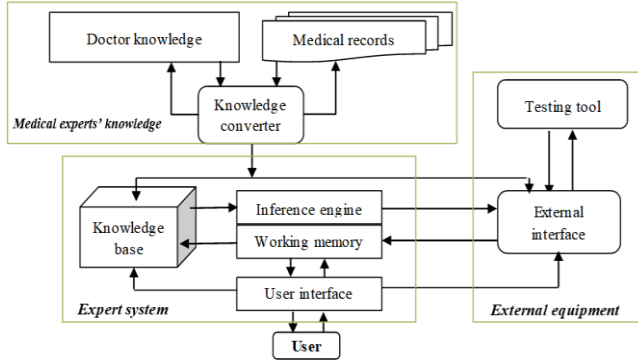


Fig. 1. Components of a typical medical expert system

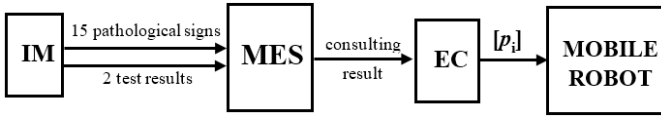


Fig. 2. Operation process of a medical expert system

A specific MES probably performs multiple tasks of acquiring knowledge, processing information, and logical inferences. The basic components of a typical MES illustrated in [1] and organized as Fig. 1 including:

- Knowledge base: convert medical experts' knowledge into rules and organized it as a database. The collection of medical experts' knowledge is performed in a systematic, selective and validated manner.
- Inference machine: evaluate rules relating to provided inputs to make decisions.
- User interface: collect input data and display consulting results.

The operational process for the typical MES is shown in Fig. 2. The first task for MES is collecting patient data: personal information and symptoms of diseases for consulting or the desired examination department through the Information module (IM). With the collected information, the MES module processes the information and offers consulting results. Then, the user chooses the appropriate examination department. The user's selection will be encoded by the encoding module (EC) into a series of points as a navigation signal for a mobile robot.

B. Design of a medical expert system

1) Program organization

The patients will directly operate on the MES interface of the mobile robot. There are 2 options on the main screen: "Consult the examination department according to symptoms" (referred to as Option 01) and "Actively choose the examination department" (referred to as Option 02). If the patient goes to the examination department for the first time and does not know where exactly to go, choose Option 01. If not, choose Option 02. The program scheme for MES is described in Fig. 3.

2) Forming the knowledge base for the MES

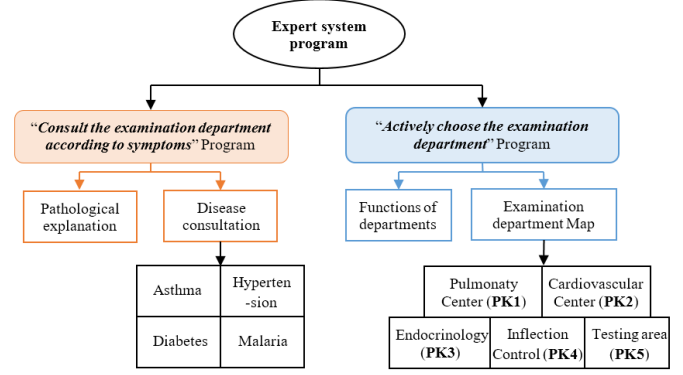


Fig. 3. Program scheme for MES

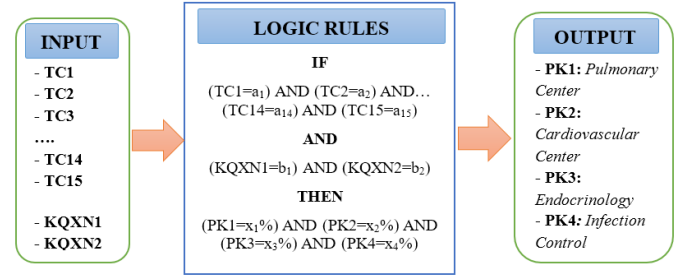


Fig. 4. Knowledge base of Medical Expert System

(KQXN1) FPG: fasting plasma glucose test
(KQXN2) RDTs: rapid diagnostic test

The database of Option 01 is collected and synthesized based on the database of Vietnam Ministry of Health [12]. The MES for Initial diagnosis includes 4 popular diseases including: Asthma, Diabetes, Hypertension and Malaria. The input set of MES contains 15 typical symptoms with 2 values of test results selected in [13], described in Table I. The inputs are linked together by logical operators and weights to form inference based on If-Then rules shown in Fig. 4. TC is symptoms, KQXN is test results and PK is the desired examination department.

3) Creating a mechanism for inferring decision-making

The mechanism of inference is carried out with 17 inputs (15 symptoms and 2 test results). The reasoning process bases on the logic code in the knowledge base of MES. Symptoms are divided into two main groups: external signs and internal feelings. Each symptom TC_i and test results $KQXN_i$ in the inference set will be assigned with a corresponding a_i or b_i called a weighted value indicating the degree of dependence of that symptom on each value of weight vary between $[0 \div 1]$ depending on "yes – no" or "mild – middle – high" choices or values index of tests results, described as follows:

- Symptom 1: Cough

$$TC_1 = \begin{cases} yes \rightarrow a_1 = 1 \\ no \rightarrow a_1 = 0 \end{cases} \quad (1)$$

- Test result 1: FPG

$$KQXN_1 = \begin{cases} < 100 \sim (mild) \rightarrow b_1 = 0.2 \\ 100 \div 125 \sim (middle) \rightarrow b_1 = 0.5 \\ > 125 \sim (high) \rightarrow b_1 = 0.8 \end{cases} \quad (2)$$

The consulting results are shown in a graph with the percentages of 4 diseases x_i (%) and displayed on the result interface to offer consultation. This result output depends on the set of input values provided by the patient. Some symptoms can be seen in many different diseases, however, when combined with different symptoms will lead to different diagnoses depending on the weighted values assigned in the logic code. The more detailed the inputs set are, the more “obvious” the results are, giving a more reliable consultation for the MES.

TABLE I. DESCRIPTION OF INPUT SYMPTOMS

No.	Symptoms	Input variable
1	Cough	TC1
2	Rhinorrhea	TC2
3	Hard to breathe	TC3
4	Shortness of breath	TC4
5	Angina (Chest pain)	TC5
6	Sweating	TC6
7	Dizziness	TC7
8	Nosebleeds	TC8
9	Fuzzy eyes	TC9
10	Nausea	TC10
11	Increased appetite	TC11
12	Quick weight lose	TC12
13	Chill	TC13
14	High fever	TC14
15	Fatigue	TC15

III. APPLICATION OF CONSULTING RESULTS FOR NAVIGATING MOBILE ROBOT

The mobile robot in this study is named “Maibot”. Maibot was designed with 4 wheels, 2 steady arms, uses the line marking sensors to find the route, and the ultrasonic sensors to detect obstacles. This study uses the general control model presented in [13] for a functional modular control solution. The control diagram of this system is shown in Fig. 5 with 3 main modules: Signal Processing (SP), Route Programming (RP) and Movement Controlling (MC).

The SP module processes the received signals to give out a series of way point p_K related to an examination department and obstacle data concerning distances $[d_i]$ in three directions and a movement command $[\varphi_i]$.

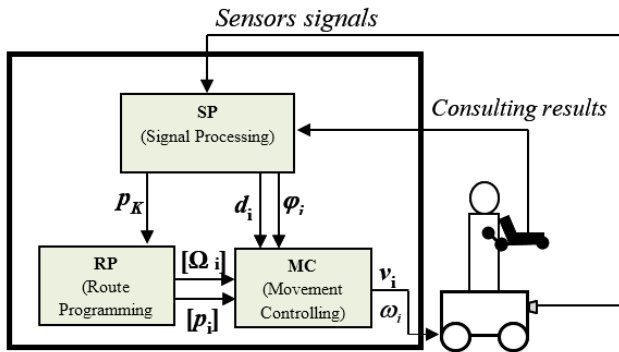


Fig. 5. Diagram of Maibot's module-based control system

The linguistic values of $[d_i]$ are “near”, “middle”, “far”. The values of obstacle direction $[\varphi_i]$ are used to control the robot avoid collision and set as follows:

$$\varphi_i = \begin{cases} 0 & \text{for "go_ahead"} \\ +90 & \text{for "turn_right"} \\ -90 & \text{for "turn_left"} \\ 180 & \text{for "go_back"} \end{cases} \quad (3)$$

The RP module calculates the route from Maibot's current point to the destination p_K . That means the output of RP is a way $[p_i]$ from the standing point to the examination department p_K . The other output of PK is the values of $[\Omega_i]$ are follows:

$$\Omega_i = \begin{cases} 0 & \text{for "go_ahead"} \\ +90 & \text{for "turn_right"} \\ -90 & \text{for "turn_left"} \\ 180 & \text{for "go_back"} \end{cases} \quad (4)$$

The MC module performs movement control according to the hierarchical behavior-based method introduced in [14]. According to the definitions in [15], bottom-layer behaviors are primitive behaviors that are simple or self-directed serving a single purpose and operate in a stimulus-responsive model.

IV. TESTING

A. Experimental mobile robot model

The research team designed a humanoid mobile robot model and named it “Maibot” (Fig. 6). Maibot was designed to resemble a medical staff serving in medical examination departments. Maibot has a 155cm height; arms 35 cm x 41 cm; stand 35 cm x 45 cm x 7 cm. On the moving stand, there are 2 motor mounts with active wheels at the rear, 2 omnidirectional wheels in front with ultrasonic sensors, and line marking sensors. Maibot wears a white blouse, skirt and has a human face, holding a laptop integrated with the MES program.



Fig. 6. Appearance of the Maibot model

The MES for examination department consultation integrated in the laptop, which is hold on the arms of Maibot. Using the consulting results, Maibot calculates the route to move from the current position to the destination and conducts

the movement to guide the patient. At each intersection, Maibot analyzes the next direction before making decisions of going straight or turning across the road markings. During the movement, the ultrasonic sensors are always active. If Maibot detects an obstacle within 50 cm, it will stop and wait for 10 seconds and create a warning signal to ask for assistance in removing the obstacle and then continue to perform the route.

B. Testing results

Maibot was tested in a simulated examination department workspace that has 7 destinations, 5 of them are general examination departments, and a specific situation: A patient comes to the examination department and be welcomed by Maibot. After the patient provides personal and disease-related information, the program offers results and advice to the appropriate examination department. Maibot then performs the navigation task.

The MES was designed with 2 main programs: “Option 01” and “Option 02”. When the patient interacts with Maibot, they will communicate with the MES through the main interface as Fig. 5, and insert personal information for recording patient data. After that, the patient can choose one of the two programs mentioned above to continue the next steps.

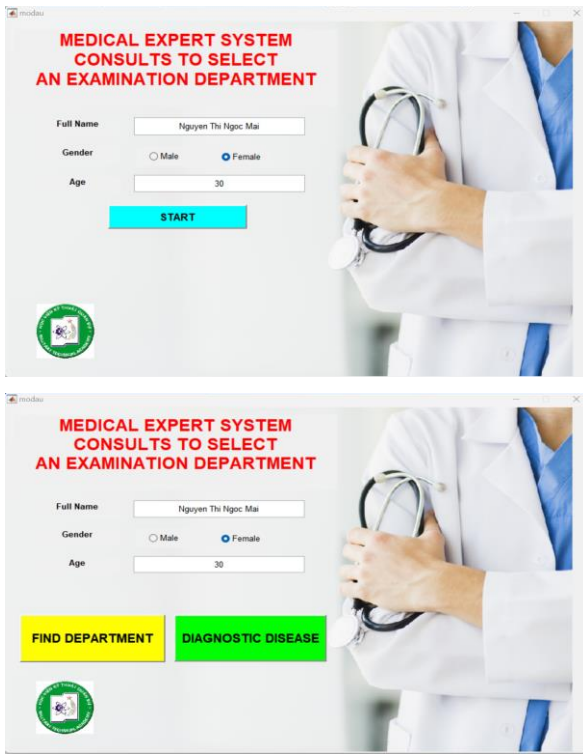


Fig. 7. Initial interface

1) The “Option 02” program

After pressing the button “Find Department on the interface Fig. 7” to select examination department by his/herself, a new interface concerning the defined map of the hospital will be given as shown in Fig. 8. The patient can select the desired examination department on the map. After selecting, the system will send the data of the selected examination department including a p_K value to the RP module. Based on

the p_K value, the RP module will calculate the route from the first position p_0 to the last position p_K as a vector $\mathbf{p}=[p_0, \dots, p_K]$. The EC module will carry out a path planning algorithm to generate a set of robot action at point(s) p_i (in the middle from p_0 to p_K) such as a motion command $\mathbf{v}_i=[v_i, \omega_i]$, where v_i is a linear velocity, and ω_i is a rotation angle at the intersection point p_i . After that, the EC module sent the set of commands $\mathbf{v}_i$ to the motion control module to guide Maibot following the given map involving straight lines (between the points) and sharp turns (at the intersection point) to approach the last position.

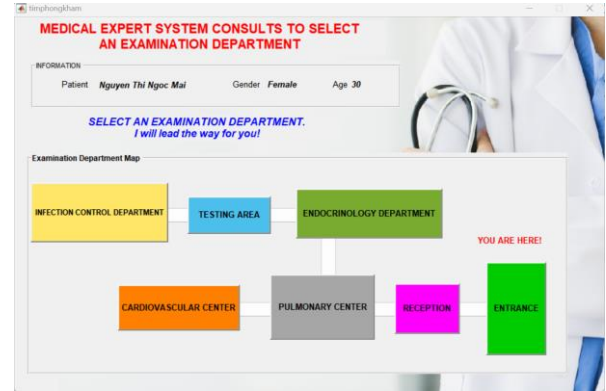


Fig. 8. Interface of “Option 02” program

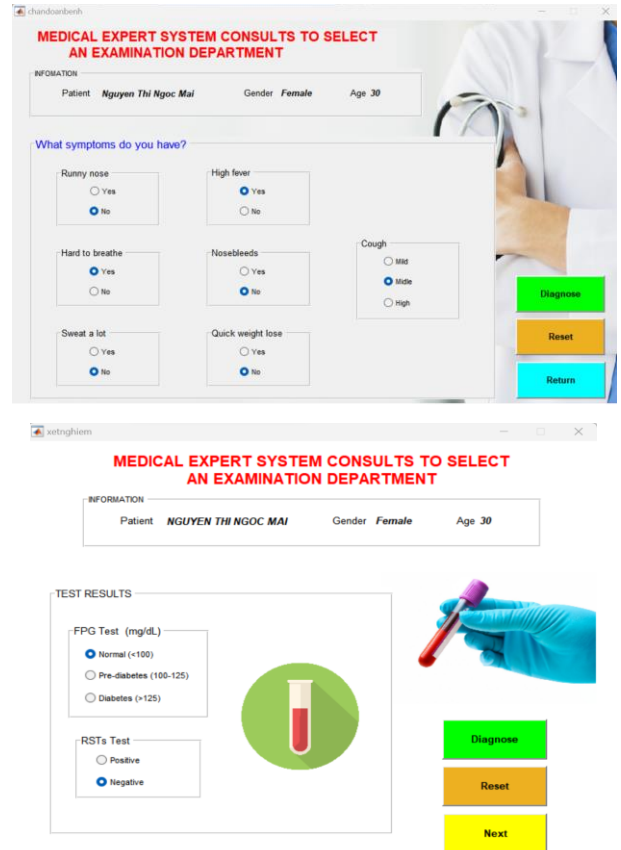


Fig. 9. Interfaces of “Option 01” program

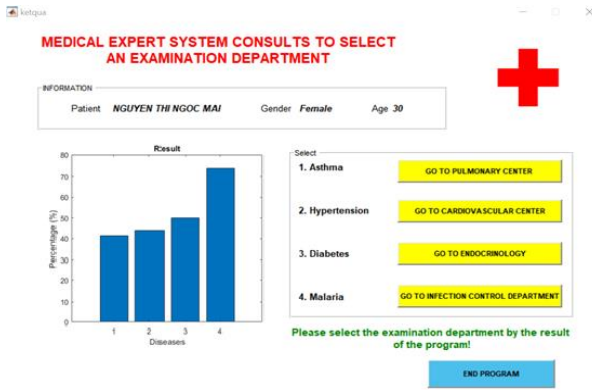


Fig. 10. Result interface

2) The “Option 01” program

The “Option 01” program is used for patients who are in doubt about their conditions and do not know the desired examination department to go to. In this situation, the patient presses the button “Diagnostic disease” on the interface Fig. 7 to be consulted an examination department according to symptoms. After, the MES offers an interface that contains several typical symptoms and test results, as in Fig. 9. The consulting result will be expressed on screen an expressed on the screen as in Fig. 10.

After the patient enters a set of symptoms in Fig. 9, the MES automatically calculates the diagnosis results and recommends the appropriate examination department on the Result interface as Fig. 10. The patient will choose the desired examination department according to the diagnosis result from the MES. At this moment, the MES will send the selection result to the SP module and this module will send a value p_K to the RP module. Based on the p_K value, the RP module calculates the moving route.

C. Evaluate the testing results

Case 1. Select the desired examination department directly on the MES: Maibot allows the patient to communicate with the MES through a mouse, keyboard, and touch screen. The patient can choose the desired examination department directly through the “Option 02” program or make the initial diagnosis with the “Option 01” program. With the “Option 01” program, the input set contains 15 symptoms and 2 test results selectable for the patient, and the MES uses this set to diagnose 4 common diseases. The diagnostic result of this program is displayed clearly with references so that the patient can make the next decision by selecting one of the desired examination departments.

Case 2. Maibot guides the patient to the desired examination department based on output of Case 1: Maibot encrypts the desired examination department location (based on Case 1) and identified the appropriate route to be used to guide the patient. During the navigation process, it uses the result of route programming to simultaneously carry out the process of detecting and avoiding obstacles. In the event of an obstruction, it stops for 10 seconds to wait for the obstacles to move away. If the obstacles are fixed on the line, Maibot will

create a warning signal for asking to remove obstacles. During the experiment, it always found the way from its current position to the desired examination department and made a stop in case of detecting obstacles in the moving process.

V. CONCLUSION

This article introduced a MES that can advise on examination department selection based on symptoms provided by the patient. The results of this MES’s consultation are used for mobile robot programming to find the path and offer a guidance for patients to the suitable examination department. Tests on a robot model in a simulated examination department workspace have been carried out and analyzed to evaluate the set purposes and efficiency.

The results showed that the mobile robot with a MES can effectively lead the patient to a suitable examination department based on their symptoms. The robot program of routine calculation always finds a path to the consulted examination department. However, the navigation is carried out only on a ground examination department without floors and stairs; limited quantity of examination departments; mobile robots only move along the road markings, the pathological situations that a MES can diagnose stop at some basic diseases.

In the near future the research team will develop the ability of this MES to advise with more pathological situations; the communicating ability among mobile robots on each floor of a workspace to relay the consultation results (instead of climbing stairs); optimize the programmed path to the desired destinations; then carry out the test in a hospital workspace to get realistic results.

REFERENCES

- [1] Mai N.A and Pham D. D. (2019) Design of a fuzzy medical expert system for consulting Prostate diseases. Proceedings of FAIR2019, pp. 102-109. DOI: 10.15625/vap.2019.00014
- [2] J. S. Aikins, J. C. Kunz, E. H. Shortliffe, and R. J. (1983) PUFF - an Expert system for interpretation of pulmonary function data. Computers in Biomedical Research, pp-199-208, vol.16
- [3] B. G. Buchanan and E. H. Shortliffe (1984) Rule-based expert systems: The Mycin experiments of the Stanford Heuristic Program Project. Addison-Wesley Publishing Co., Inc
- [4] Mai N. T. N., Chung T. Q., Dat D. T., Mai N.A (2018) Design of a Medical Expert System for Consulting Tuberculosis Diagnosis in Vietnam Rural Areas. IFMBE Proceedings (BME7), pp. 577-583.
- [5] Abu Ghali, M. J., Mukhaimer, M. N., Abu Yousef, M. K., & Abu Naser, S. S. (2017). Expert System for Problems of Teeth and Gums. International Journal of Engineering and Information Systems (IJEAIS), 1(4), 71-88
- [6] Spyros G. Tzafestas (2014) *Introduction to Mobile Robot Control*. First edition, p. 1
- [7] Hung N.C., , Ha X. N, Anh M. N, Lam D. B, and Hai M. P, *A Modular Design Process for Developing Humanoid Mobile Robot Viebot*, Advances in Science, Technology and Engineering Systems Journal Vol. 3, No. 4, 230-235 (2018)
- [8] Gil P. S, Yuko Y., Hirokazu I., et. al, (2022), *Robots and Robotics in Nursing*, Journal of Healthcare 10(8), DOI: 10.3390/healthcare10081571
- [9] Patrick H. and Sven W., (2014), The Receptionist Robot in the, International Conference on Human-Robot Interaction, DOI: 10.1145/2559636.2559784
- [10] Christopher P. L. and Valérie J., (2021), *Mobile Remote Presence Robots for Medical Consultation and Social Connectedness*, In book: Public Health and Informatics, DOI: 10.3233/SHTI210328

- [11] Zheng W., Sicong L., Jing P., and Michael Z. C., (2018), The Next-Generation Surgical Robots, In book: Surgical Robotics, DOI: 10.5772/67515
- [12] Vietnam Ministry of Health (2014), Professional documents guiding medical examination and treatment at commune and ward health stations. Issued under Decision No. 2919/QD-BYT
- [13] Mai, N.A. and Janschek, K. (2012) *Generic system architecture for behavior-based mobile robot control using fuzzy logic*. International Conference on Control, Automation and Information Sciences, ICCAIS 2012, DOI: 10.1109/ICCAIS.2012.6466598
- [14] Mai, N.A. and Janschek, K. (2012) Performance results of qualitative optical flow processing integrated in fuzzy logic-based behavioral mobile robot control. 10th IFAC Symposium on Robot Control, Vol. 45, Iss. 22, pp. 895-901
- [15] Mai, N.A. and Janschek, K. (2009) Bio-inspired Optical Flow Interpretation with Fuzzy Logic for Behavior-Based Robot Control. RAAD, Corpus ID: 16833041.